

PROFILE

- A Ph.D. student in Electrical and Computer Engineering exploring Memory Systems, Operating Systems, and Storage Systems
- Gaining experience designing and optimizing computer architectures and systems, with a particular interest in **energy-efficient system design** and **AI system design**
- Seeking **postdoc and internship opportunities** in computer architecture and systems to contribute to innovative projects and expand technical skills

EDUCATION

Ph.D. in Electrical and Computer Engineering **Fall 2021 Year - Spring/Fall 2027 (expected)**
Syracuse University

Bachelor of Science in Electrical and Computer Engineering **Fall 2017 - Spring 2021**
Western Washington University *GPA: 3.96/4.0 | IEEE-Eta Kappa Nu*

PUBLICATIONS

- **Shao-Peng Yang**, Sandy Lin, Gongjin Sun, Dongyang Li, Shuyi Pei, Bryan Kim. Rethinking Near-Data Processing Execution: Enabling Device Collaboration with CXL Peer-to-Peer Communication. To appear in *IEEE Cluster*, 2026
- Hanqiu Chen, **Shao-Peng Yang (co-first author)**, Mohammadreza Soltaniyeh, Shuyi Pei, Andrew Chang, Bryan S. Kim, Cong Hao. COMETS: Cost-effective Multi-node Efficient Training System with Memory Pooling and Sharing. *ACM International Conference on Supercomputing (ICS)*, 2026
- **Shao-Peng Yang**, Minjae Kim, Sanghyun Nam, Juhung Park, Jin-yong Choi, Eeye Hyun Nam, Eunji Lee, Sungjin Lee, Bryan S. Kim. Overcoming the Memory Wall with CXL-enabled SSDs. *USENIX Annual Technical Conference (ATC)*, 2023

EXPERIENCE

Research Assistant **Fall 2021 - Present**
Syracuse University
Advisor: Dr. Bryan S. Kim

- **Energy-Efficient CXL Memory Systems** (Spring 2026 – present)
 - Investigating potential energy inefficiency in CXL memory tiering systems with LPDDR
 - Profiling energy benefits of LPDDR-based CXL memory using gem5
 - Developing energy-aware tiering policies to address the performance-only focus of existing systems (TPP, Mementis, Colloid)
- **Optimizing performance for CXL-based NDP systems** (Spring 2025 – Spring 2026)
 - Studied new execution models for CXL-based NDP systems with peer-to-peer (P2P) communication
 - Proposed near-data processing with collaborative peers (NDP-CP) to break the assumption that NDP devices operate independently
 - Implemented and evaluated the system on a NUMA-based emulation platform
- **CXL integration to LLM training system** (Spring 2024 - Spring 2025)
 - Studied the state-of-the-art LLM training systems and techniques such as 3D parallelism and ZeRO
 - Designed a CXL-based system utilizing memory pooling and sharing to improve the performance and the cost efficiency for LLM training
 - Built a simulator in Python to estimate training performance for a large GPU system with integration of CXL devices
- **Memory management in CXL fabric architecture** (Fall 2023)
 - Explored the design challenges of the CXL fabric manager for achieving high system performance and balancing device loads

- Researched CXL memory management in terms of allocation, translation, and protection
- **Design of CXL-flash** (Fall 2022 - Spring 2023)
 - Investigated the design challenges for creating a novel memory expansion device using CXL and flash memory
 - Applied techniques such as caching and prefetching to hide the high latency of flash memory in the CXL-flash
 - Built a trace-driven simulator in C++ for testing the CXL-flash design

Intern, Memory Solutions Research Engineer

May 2025 - August 2025

Samsung Semiconductor, Inc.

Supervisor: Caroline Kahn

- **CMM-H for LLM Inference**
 - Explored the benefits of CMM-H for CPU-based multi-turn LLM inference
 - Studied how CMM-H can store KV cache for long sequences and model weights
- **CMM-H and Data Trimming**
 - Identified a limitation of CMM-H as memory, where data is not invalidated after use
 - Showcased performance overheads due to high write amplification when CMM-H flash memory becomes "full", as stale data is not invalidated

SKILLS

Programming C++, Parallel and Multithreaded Programming, C, Assembly, Python, Matlab

Hardware Design Verilog, Altium, Digital Logic Design, skills in utilizing lab tools such as oscilloscopes